

A2S-SFT

Audio Perception
Training



Audio Encoder & Aligner

Sentence Recon.
Training



Language Model

Joint Fine-
Tuning



All Para.

acoustic-to-semantic

q

Policy
Model



O1

O2

...

O_N

Recov-Recon
Dynamic Reward

Simple Reword
function

R1

R2

...

R_N

Group
Compute

A1

A2

A_N

O1

Mega-ASR is
a model for
Speech rec...

23.5%

O2

...

O_N

WER

< 50%
 $\alpha_{recover}$

> 50%
 α_{recon}

Word-level Reward

O_K

GT

Levenshtein ratio

$$R_{word} = \frac{n_C}{n_C + w_h \cdot n_{hard} + w_s \cdot n_{soft} + 1e-8}$$

Sent. Level Reward

LCS Recall Reward

→ R

Length re

→ R